

Antifolate / nucleoside: 7 molecules stopped early, all small molecules

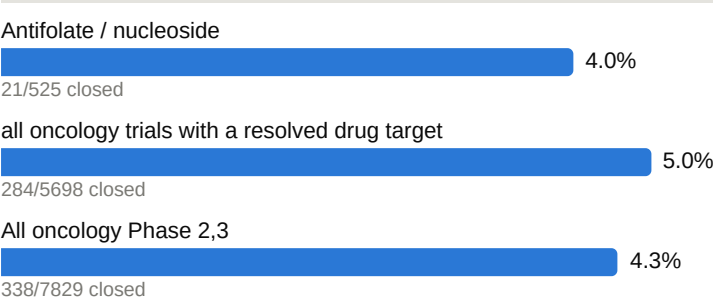
21 stopped trials, but 7 molecules — all small molecules: Gemcitabine, Pemetrexed, Fluorouracil, Methotrexate and 3 more. Against 5.0% for all oncology trials with a resolved drug target and 4.3% across all oncology Phase 2,3 trials in the same window. The rate counts trials that stopped early for efficacy, safety or benefit–risk reasons. Trials that ran to completion and missed their primary endpoint are counted separately below, never in the rate.

4.0%	13 / 7 / 1	31.2%	1.1%
21 of 525 closed trials (95% CI 2.6%–6.0%)	efficacy / safety / both (adds to 21)	of 1682 trials have closed · 1157 still open or unresolved	stopped for a biological reason within 3 years of starting (95% CI 0.6%–2.0%; 696 still at risk; 1.5% area-wide)

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Completed and missed the primary endpoint. Of 351 completed trials in this segment, 10 posted a primary result on ClinicalTrials.gov that can be read — the sponsor's own comparison held to the sponsor's own threshold, or its statement that the endpoint was missed: 6 missed and 3 met, 1 split across co-primary endpoints. Counted separately and never in the rate above; a floor, not a rate. 180 of the 320 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

How this compares



What is in this class: 553 distinct experimental drugs across 1682 trials (small molecule 1682, antibody 756, unknown 340, protein 131). Most tested: GEMCITABINE (502), CAPECITABINE (440), OXALIPLATIN (395), FLUOROURACIL (393), PACLITAXEL (301). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 21 stops came from 21 sponsor-asset programmes across 21 sponsors; the largest (Universität Duisburg-Essen) contributed 1; removing that programme's trials from both sides leaves 3.8%. Screened alongside 85 other segments in this area, it does not survive a 10% false-discovery correction ($q=1$, Benjamini-Yekutieli). A further 44 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 12.4%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2015–2017	12	239	5.0%	2.9%–8.6%
2018–2020	5	173	2.9%	1.2%–6.6%
2021–2024	4	113	3.5%	1.4%–8.7%

Where the stops sit

Efficacy / futility: 13, Safety: 7, Efficacy and safety: 1.
Sponsors with most stops: University of Nebraska (1), BerGenBio ASA (1), Tufts Medical Center (1), Roswell Park Cancer Institute (1), Ukrainian Society of Clinical Oncology (1), AHS Cancer Control Alberta (1).

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT04141995	2	University of Nebraska	DIGOXIN, FLUOROURACIL, CALCIUM, IRINOTECAN	2025-05	Efficacy	Due to low accrual and futility
NCT05469178	2	BerGenBio ASA	BEMCENTINIB, PEMBROLIZUMAB, PEMETREXED, CARBOPLATIN	2025-04	Efficacy	Lack of efficacy
NCT04856176	2	Tufts Medical Center	Granulocyte-Macrophage Colony-Sti..., PEMBROLIZUMAB, PEMETREXED, PACLITAXEL	2024-12	Efficacy	Futility
NCT03333486	2	Roswell Park Cancer Institute	CYCLOPHOSPHAMIDE ANHYDROUS, FLUDARABINE PHOSPHATE	2023-08	Efficacy	Futility analysis determined closure
NCT04937738	2	Ukrainian Society of Clinical Oncology	DOCETAXEL ANHYDROUS, OXALIPLATIN, LEUCOVORIN, FLUOROURACIL	2023-02	Efficacy	High rate of disease progression in XELOX arm
NCT02295722	2	AHS Cancer Control Alberta	GEMCITABINE, MELPHALAN	2023-02	Efficacy	It did not show a significant benefit to justify completing the full target accrual.

The 6 most recent of 21 stopped trials. Every molecule behind all 21 is named above; the complete trial list, the trials still running and the terminations with no readable cause ship with the dataset and the evidence package.

Method

Denominator: ClinicalTrials.gov interventional Phase 2,3 oncology trials started 2015–2024 that have closed (completed or terminated).
Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 76% of industry oncology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

Limits

Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 31.2% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

The whole cohort, trial by trial. The evidence package names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. €99, delivered immediately: clinicaltrialfailures.com/asset-check

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.